INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA
SALTLAKE
(School of University of Engineering and Management, Kolkata)

Degree: BCA, Stream: BCA

Term – II , First Semester Examination, October- 2025

Subject Code: BCACC102

Subject Name: Digital Electronics

Full Marks: 30 Duration: 1 Hour

Part - A
Answer any 5 questions out of 6
Each question carries 2 marks (2 × 5)

1. Why do we prefer K-map simplification over Boolean algebra for larger expressions? 2
2. Why is DeMorgan's theorem important in digital circuit design? 2
3. What is the main difference between a multiplexer (MUX) and a decoder? 2
4. Why are don't care conditions used in Karnaugh Maps (K-maps)? 2
5. What is the difference between registers and counters? 2
6. What is race around condition and how it is solved? 2

Part - B
Answer any 2 questions out of 3
Each question carries 5 marks (5 × 2)

7. Describe the steps required to convert SR flipflop to JK flipflop. 5
8. Implement 8x1 Multiplexer using 4x1 Multiplexer and some additional gates. 5
9. Design a MOD-3 synchronous UP counter using JK flipflop and explain the working. 5

Part - C
Answer any 1 question out of 2
Each question carries 10 marks (10 × 1)

10. Implement 4x16 Decoder using 2x4 decoders only 10
11. Design a BCD to Gray code converter circuit. 10

--End--